James Stevenson
http://jsstevenson.github.io
Email : james.sharpsteen@gmail.com

Experience

Nationwide Children's Hospital *Sep 2020 - Present*
Bioinformatics Software Developer

- Lead development of biomedical concept normalizer software, including Thera-Py and Disease Normalizer. Use Python to perform rapid data ingest methods over millions of records from a variety of biomedical data sources and formats into DynamoDB, PostgreSQL, and Neo4j backends.
- Contribute full-stack development to Drug-Gene Interaction Database and Fusion Builder applications, from data architecture planning to UX/UI evaluation, using Ruby on Rails, Python, and Javascript/Typescript. Provide analyses and produce figures for manuscripts and working papers.
- Develop Rust-based interface to genomic annotation liftover tooling to support variation analysis workflows. Achieved 80% reduction in startup time over existing solution.
- Help define NIH standards for AI-ready biomedical data as a Bridge2AI Standards Core member. Craft expectations for data quality and provenance, workflow reproduceability, and protecting fairness/identifying ethical concerns.

Education

Georgia Institute of Technology *May 2024*
Master of Science in Computer Science (Specialization: Machine Learning)

University of Puget Sound *May 2013*
Bachelor of Arts in Philosophy and Politics & Government

Publications

Matthew Cannon, Wesley Goar, In-Hee Lee, **James Stevenson**, Amy Heiser, Nathan Sheffield, James Eddy, Monica Munoz-Torres, Sek Wong Kong, Alex H Wagner. "Bridge2AI Recommendations for AI-Ready Genomic Data." *Bioinformatics Advances*, Volume 6, Issue 1, 25 October 2026. DOI: 10.1093/bioadv/vbag305.

Jeremy A. Arbesfeld, Estelle Y. Da, **James S. Stevenson**, Kori Kuzma, Anika Paul, Tierra Farris, Benjamin J. Capodanno, Sally B. Grindstaff, Kevin Riehle, Nuno Saraiva-Agostinho, Jordan F. Safer, Jonathan Casper, Maximilian Haeussler, Aleksandar Milosavljevic, Julia Foreman, Helen V. Firth, Sarah E. Hunt, Sumaiya Iqbal, Melissa S. Cline, Alan F. Rubin, Alex H. Wagner. "Mapping MAVE data for use in human genomics applications". *Genome Biology*, Volume 26, Number 179, 25 June 2025. DOI: 10.1186/s13059-025-03647-x.

Matthew Cannon, **James Stevenson**, Kathryn Stahl, Rohit Basu, Adam Coffman, Susanna Kiwala, Joshua F McMichael, Kori Kuzma, Dorian Morrissey, Kelsy Cotto, Elaine R Mardis, Obi L Griffith, Malachi Griffith, Alex H Wagner. "DGIdb 5.0: rebuilding the drug–gene interaction database for precision medicine and drug discovery platforms." *Nucleic Acids Research*, Volume 52, Issue D1, 5 January 2024. DOI: 10.1093/nar/gkad1040.

Matthew Cannon, **James Stevenson**, Kori Kuzma, Susanna Kiwala, Jeremy L Warner, Obi L Griffith, Malachi Griffith, Alex H Wagner. "Normalization of Drug and Therapeutic Concepts with Thera-Py." *JAMIA Open*, Volume 7, Number 4, 2023. DOI: 10.1093/jamiaopen/ooad093.

Preprints

Daniel Puthawala, Brendan Reardon, Lawrence Babb, Kori Kuzma, **James S. Stevenson**, Wesley A. Goar, Robert H. Dolin, Robert R. Freimuth, Catherine Procknow, Beth Pitel, Parijat Kundu, Akash Rampersad, Eliezer Van Allen, Alex H. Wagner. "The GA4GH Categorical Variation Representation Specification: A Unified Computational Framework for Reasoning over Genomic Variant Categories." *bioRxiv*, 12 February 2026. DOI: 10.64898/2026.02.10.705161.

Matthew Cannon, Anastasia Bratulin, **James Stevenson**, Kathryn Perry, Adam Coffman, Susanna Kiwala, Lars Schimmelpfennig, Heather Costello, Joshua F McMichael, Malachi Griffith, Obi L. Griffith, Alex H. Wagner. "dgiLIT: A Method for Prioritization and AI Curation of Drug-Gene Interactions." *bioRxiv*, 19 January 2026. DOI: 10.64898/2026.01.16.699733.

Wesley A. Goar, Daniel Puthawala, Kori Kuzma, Anastasia Bratulin, Austin A. Antoniou, Jeremy A. Arbesfeld, Lawrence Babb, Kyle Ferriter, Terry O'Neill, **James S. Stevenson**, Kathryn Perry, Matthew Cannon, Jiachen Liu, Xuelu Liu, Brian Walsh, Savanna Funk, William C. Ray, Bimal P. Chaudhari, Heidi L. Rehm, Alex H. Wagner. "The Clinical Genomic Variation Landscape." *medRxiv*, 18 December 2025. DOI: 10.1101/2025.11.04.25339115.

Timothy Clark, Harry Caufield, Jillian A. Parker, Sadnan Al Manir, Edilberto Amorim, James Eddy, Nayoon Gim, Brian Gow, Wesley Goar, Melissa Haendel, Jan N. Hansen, Nomi Harris, Henning Hermjakob, Marcin Joachimiak, Gianna Jordan, In-Hee Lee, Shannon K. McWeeney, Camille Nebeker, Milen Nikolov, Jamie Shaffer, Nathan Sheffield, Gloria Sheynkman, **James Stevenson**, Jake Y. Chen, Chris Mungall, Alex Wagner, Sek Won Kong, Satrajit S. Ghosh, Bhavesh Patel, Andrew Williams, Monica C. Munoz-Torres, the Bridge to Artificial Intelligence (Bridge2AI) Consortium members. "AI-readiness for Biomedical Data: Bridge2AI Recommendations." *bioRxiv*, 24 November 2024. DOI: 10.1101/2024.10.23.619844.

Conference Presentations

Cancer Genomics Consortium Annual Meeting, St. Louis, MO *Aug 2023*
"Overcoming challenges in semantic alignment of therapeutics knowledge using TheraPy"

- Abstract selected for Technologist Travel Award

CASCAID Regional Conference, Bellingham, WA *Aug 2012*
"You Eat What You Are: Deconstruction, Dialectics, and Digestion"

- Selected for Best Student Paper Award

Service

Open Biological and Biomedical Ontology Foundry *Mar 2024 - present*
Technical Working Group Operations Liaison

Teaching

Instructional Aide (CS7650: Human-Computer Interaction) *Jan 2021 - Aug 2021*
Georgia Institute of Technology

Instructor (COMM292: Intercollegiate Debate) *Sep 2013 - May 2015*
University of Puget Sound